

ON NEGATIVE ESTIMATES OF VARIANCE COMPONENTS UNDER FINITE POPULATION MODELS

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SUMMARY

The customary analysis of variance models with random components are considered from the point of view of finite population sampling theory. It is shown that under the finite model the usual analysis of variance estimators are no longer unbiased for estimating variance components, but estimate quantities which have a well-defined possibility of taking negative values. The results are illustrated with appropriate numerical examples.

1. Introduction

The variance components models are appropriate to many types of data arising in industrial experiments. The customary estimation procedure based on the analysis of variance consists of equating observed mean squares to expected values and then obtaining linear functions of the mean squares that are unbiased for the variance components. For balanced designs the estimators of the variance components thus obtained possess certain optimal properties. (See Graybill (1954), Graybill and Wortham (1956), Graybill and Hultquist (1961)). However, the analysis of variance estimators can produce negative estimates for variance components which are by definition non-negative. This problem has been considered by several authors and possible causes have been put forward for these negative estimates. One of these causes is that the usual assumption of non-correlated identically distributed random variables is incorrect. This can arise when the customary infinite model may erroneously be applied in the case of a finite model. McHugh and Mielke (1968) have shown that under a finite model situation of the balanced one-way random model, the usual analysis of variance estimator of the between-variance component is no longer unbiased, but estimates a quantity which can take on negative value. In this paper, these results are extended to cover some higher nested and crossed classification random effects models.

2. Nested Random Effects Model

The nested design involves successive random selection from a population with two

or more factors each nested within the previous. Consider the case of a four-stage balanced nested design with factors A, B, C, and D; in which D is nested in C, C is nested in B, and B is nested in A.

2.1 The Infinite Case

The customary variance components model for the above design is

$$(1) \quad y_{ijkl} = \mu + a_i + b_{j(i)} + c_{k(ij)} + d_{l(ijk)}$$

($i=1, \dots, n_a$; $j=1, \dots, n_b$; $k=1, \dots, n_c$; $l=1, \dots, n_d$), where μ is the population mean assumed to be fixed, and $a_i, b_{j(i)}, c_{k(ij)}$, and $d_{l(ijk)}$ are random effects corresponding to the factors A, B, C, and D; and are assumed to be independently distributed of themselves and each other with zero means and respective variances $\sigma_a^2, \sigma_b^2, \sigma_c^2$, and σ_d^2 . The model (1) under the above assumptions is called the "infinite model". The expected mean squares of this model are presented in Table 1. (See Anderson and Bancroft (1952, p. 325), Bennett and Franklin (1954, p. 403)).

TABLE 1
Mean Squares and Expected Mean Squares for the
Infinite Model

Source	Mean Squares	Expected Mean Squares
A	S_a^2	$\sigma_d^2 + n_d \sigma_c^2 + n_c n_d \sigma_b^2 + n_b n_c n_d \sigma_a^2$
B in A	S_b^2	$\sigma_d^2 + n_d \sigma_c^2 + n_c n_d \sigma_b^2$
C in B	S_c^2	$\sigma_d^2 + n_d \sigma_c^2$
D in C	S_d^2	σ_d^2

The usual analysis of variance estimators of the variance components obtained from Table 1 are

$$(2) \quad \tilde{\sigma}_d^2 = S_d^2, \quad \tilde{\sigma}_c^2 = (S_c^2 - S_d^2)/n_d,$$

$$\tilde{\sigma}_b^2 = (S_b^2 - S_c^2)/n_c n_d, \text{ and } \tilde{\sigma}_a^2 = (S_a^2 - S_b^2)/n_b n_c n_d.$$

2.2 The Finite Case

The finite model in the case of the four-stage nested design assumes that a

population of N elements can be partitioned as follows: (i) into N_a first-stage units according to factor A; (ii) with each first-stage unit being subdivided into N_b second-stage units according to factor B; (iii) with each second-stage unit being subdivided into N_c third-stage units according to factor C; and (iv) finally each third-stage unit being subdivided into N_d fourth-stage units according to factor D, which are the ultimate elements of the population. Thus $N = N_a \cdot N_b \cdot N_c \cdot N_d$.

Let Y_{IJKL} ($I = 1, \dots, N_a$; $J = 1, \dots, N_b$; $K = 1, \dots, N_c$; $L = 1, \dots, N_d$) be the $IJKL$ -th member of the population. The population means are then

$$\mu_{IJK} = \frac{1}{N_d} \sum_{L=1}^{N_d} Y_{IJKL}, \quad \mu_{IJ} = \frac{1}{N_c} \sum_{K=1}^{N_c} \mu_{IJK},$$

$$\mu_I = \frac{1}{N_b} \sum_{J=1}^{N_b} \mu_{IJ}, \quad \text{and} \quad \mu = \frac{1}{N_a} \sum_{I=1}^{N_a} \mu_I.$$

The components of variance corresponding to the factors A, B, C and D are defined by

$$\sigma_A^2 = \frac{1}{N_a} \sum_{I=1}^{N_a} (\mu_I - \mu)^2 / (N_a - 1), \quad \sigma_B^2 = \frac{1}{N_a} \sum_{J=1}^{N_b} \sum_{I=1}^{N_a} (\mu_{IJ} - \mu_I)^2 / (N_a(N_b - 1)),$$

$$\sigma_C^2 = \frac{1}{N_a} \sum_{I=1}^{N_a} \sum_{J=1}^{N_b} \sum_{K=1}^{N_c} (\mu_{IJK} - \mu_{IJ})^2 / (N_a N_b (N_c - 1)),$$

and

$$\sigma_D^2 = \frac{1}{N_a} \sum_{I=1}^{N_a} \sum_{J=1}^{N_b} \sum_{K=1}^{N_c} \sum_{L=1}^{N_d} (Y_{IJKL} - \mu_{IJK})^2 / (N_a N_b N_c (N_d - 1)).$$

The usual sampling procedure then is to select randomly, n_a out of N_a first-stage units, n_b out of N_b second-stage units from the chosen first-stage units, n_c out of N_c third-stage units from the chosen second-stage units, and finally a random sample of size n_d out of N_d fourth-stage units from the chosen third-stage units.

The model equation in this case is the same as (1), that is,

$$y_{ijkl} = \mu + a_i + b_{j(i)} + c_{k(ij)} + d_{l(ijk)}.$$

As before, the random effects a_i , $b_{j(i)}$, $c_{k(ij)}$, and $d_{l(ijk)}$ have zero means and are independent of each other. However, they are statistically dependent among themselves with the variance-covariance structure given in Searle and Fawcett (1970). The expected mean squares for this model are shown in Table 2. (See Bennett and Franklin (1954, p. 404), Cornfield and Tukey (1956), Searle and Fawcett (1970)).

TABLE 2
Mean Squares and Expected Mean Squares for the
Finite Model

Source	Mean Square	Expected Mean Square
A	S_a^2	$(1 - \frac{n_d}{N_d})\sigma_D^2 + n_d(1 - \frac{n_c}{N_c})\sigma_C^2 + n_c n_d(1 - \frac{n_b}{N_b})\sigma_B^2 + n_b n_c n_d \sigma_A^2$
B in A	S_b^2	$(1 - \frac{n_d}{N_d})\sigma_D^2 + n_d(1 - \frac{n_c}{N_c})\sigma_C^2 + n_c n_d \sigma_B^2$
C in B	S_c^2	$(1 - \frac{n_d}{N_d})\sigma_D^2 + n_d \sigma_C^2$
D in C	S_d^2	σ_D^2

TABLE 3
A Finite Four-Stage Nested Population with
Negative $E(\tilde{\sigma}_a^2)$

		Population Elements Y_{IJKL}					
		1		2		3	
		1	2	1	2	1	2
J	I						
	K						
	L	1	2	1	2	1	2
1	1	211	71	196	96	200	40
	2	215	98	196	10	221	29
2	1	104	25	104	14	96	28
	2	121	15	65	12	84	40
3	1	29	8	15	5	40	20
	2	50	9	9	11	30	15

Suppose the usual analysis of variance estimators are calculated under the infinite model while the underlying population is finite and thus the sampling at various stages of selection is that of sampling without replacement. Then the expectations of the variance

components estimators (2) are

$$E(\tilde{\sigma}_d^2) = \sigma_D^2, E(\tilde{\sigma}_c^2) = \sigma_C^2 - N_d^{-1} \sigma_D^2,$$

$$E(\tilde{\sigma}_b^2) = \sigma_B^2 - N_c^{-1} \sigma_C^2, \text{ and } E(\tilde{\sigma}_a^2) = \sigma_A^2 - N_b^{-1} \sigma_B^2.$$

Thus if the assumptions of an infinite model are erroneously applied to a finite model situation of the balanced four-stage nested design, then the usual analysis of variance estimators $\tilde{\sigma}_a^2$, $\tilde{\sigma}_b^2$, and $\tilde{\sigma}_c^2$ are no longer unbiased and do not necessarily have non-negative expectations. This is illustrated by a numerical example. Table 3 presents a population for which $E(\tilde{\sigma}_a^2)$ is negative.

Here $N_a = 3$, $N_b = 3$, $N_c = 2$, and $N_d = 2$; and the variance components are

$$\sigma_A^2 = \frac{1}{2} \sum_{I=1}^3 (\mu_I - \mu)^2 = 94.28,$$

$$\sigma_B^2 = \frac{1}{6} \sum_{I=1}^3 \sum_{J=1}^3 (\mu_{IJ} - \mu_I)^2 = 3\,209.30,$$

$$\sigma_C^2 = \frac{1}{9} \sum_{I=1}^3 \sum_{J=1}^3 \sum_{K=1}^2 (\mu_{IJK} - \mu_{IJ})^2 = 4\,701.11,$$

and

$$\sigma_D^2 = \frac{1}{18} \sum_{I=1}^3 \sum_{J=1}^3 \sum_{K=1}^2 \sum_{L=1}^2 (Y_{IJKL} - \mu_{IJK})^2 = 323.44.$$

The expected values of the variance components estimators (2) are

$$E(\tilde{\sigma}_d^2) = \sigma_D^2 = 323.44, \quad E(\tilde{\sigma}_c^2) = \sigma_C^2 - \frac{1}{2} \sigma_D^2 = 4\,539.39,$$

$$E(\tilde{\sigma}_b^2) = \sigma_B^2 - \frac{1}{2} \sigma_C^2 = 858.74, \text{ and } E(\tilde{\sigma}_a^2) = \sigma_A^2 - \frac{1}{3} \sigma_B^2 = -975.49.$$

In general $E(\tilde{\sigma}_c^2) < 0$ if $\sigma_D^2 > N_d \sigma_C^2$,

$$E(\tilde{\sigma}_b^2) < 0 \text{ if } \sigma_C^2 > N_c \sigma_B^2, \text{ and}$$

$$E(\tilde{\sigma}_a^2) < 0 \text{ if } \sigma_B^2 > N_b \sigma_A^2.$$

3. Crossed Classification Random Effects Model

The two-way crossed classification design involves two factors, say A and B, such

that the elements of a population can be classified into classes according to both factors. The factors A and B are then said to be crossed with each other.

3.1 The Infinite Case

The customary variance components model for the above design with n_r observations per cell is

$$(3) \quad y_{ijk} = \mu + a_i + b_j + (ab)_{ij} + e_{ijk}$$

($i=1, \dots, n_a$; $j=1, \dots, n_b$; $k=1, \dots, n_r$), where μ is the population mean assumed to be fixed, and a_i , b_j , $(ab)_{ij}$, and e_{ijk} are random effects corresponding to factors A, B, interaction AB, and the residual; and are assumed to be independently distributed of themselves and each other with zero means and respective variances σ_a^2 , σ_b^2 , σ_{ab}^2 , and σ_r^2 . The model (3) under the above assumptions is called the "infinite" model. The expected mean squares of this model are presented in Table 4. (See Anderson and Bancroft (1952, p. 316), Scheffé (1959, p. 244), Steel and Torrie (1960, p. 144)).

TABLE 4
Mean Squares and Expected Mean Squares for
the Infinite Model

Source	Mean Squares	Expected Mean Squares
A	S_a^2	$\sigma_r^2 + n_r \sigma_{ab}^2 + n_b n_r \sigma_a^2$
B	S_b^2	$\sigma_r^2 + n_r \sigma_{ab}^2 + n_a n_r \sigma_b^2$
AB	S_{ab}^2	$\sigma_r^2 + n_r \sigma_{ab}^2$
Residual	S_r^2	σ_r^2

The usual analysis of variance estimators of the variance components obtained from Table 4 are

$$(4) \quad \tilde{\sigma}_r^2 = S_r^2, \tilde{\sigma}_{ab}^2 = (S_{ab}^2 - S_r^2)/n_r,$$

$$\tilde{\sigma}_b^2 = (S_b^2 - S_{ab}^2)/n_a n_r, \text{ and } \tilde{\sigma}_a^2 = (S_a^2 - S_{ab}^2)/n_b n_r.$$

3.2 Finite Case

The finite model for the case of the two-way crossed classification design assumes

that there is a population of a finite number of elements. These elements are classified into N_a classes according to factor A and into N_b classes according to factor B. Each of the $N_a N_b$ pigeonholes (cells) thus formed contains N_r elements. There are thus $N = N_a N_b N_r$ elements in the population.

Let Y_{IJK} ($I=1, \dots, N_a$; $J=1, \dots, N_b$; $K=1, \dots, N_r$) be an element belonging to the I -th class of factor A, J -th class of factor B, and the K -th element of the (I, J) -th cell. The mean of the (I, J) -th cell is

$$\mu_{IJ} = \frac{1}{N_r} \sum_{K=1}^{N_r} Y_{IJK} / N_r.$$

The means of the I -th class of factor A and the J -th class of factor B are

$$\mu_I = \frac{1}{N_b} \sum_{J=1}^{N_b} \mu_{IJ} / N_b, \quad \mu_J = \frac{1}{N_a} \sum_{I=1}^{N_a} \mu_{IJ} / N_a.$$

The overall population mean is

$$\mu = \frac{1}{N_a} \sum_{I=1}^{N_a} \frac{1}{N_b} \sum_{J=1}^{N_b} \mu_{IJ} = \frac{1}{N_a} \sum_{I=1}^{N_a} \mu_I / N_a = \frac{1}{N_b} \sum_{J=1}^{N_b} \mu_J / N_b.$$

The components of variance corresponding to the factors A, B, interaction AB, and the residual are defined by

$$\sigma_A^2 = \frac{1}{N_a} \sum_{I=1}^{N_a} (\mu_I - \mu)^2 / (N_a - 1), \quad \sigma_B^2 = \frac{1}{N_b} \sum_{J=1}^{N_b} (\mu_J - \mu)^2 / (N_b - 1),$$

$$\sigma_{AB}^2 = \frac{1}{N_a} \sum_{I=1}^{N_a} \frac{1}{N_b} \sum_{J=1}^{N_b} (\mu_{IJ} - \mu_I - \mu_J + \mu)^2 / (N_a - 1)(N_b - 1),$$

and

$$\sigma_R^2 = \frac{1}{N_a} \sum_{I=1}^{N_a} \frac{1}{N_b} \sum_{J=1}^{N_b} \frac{1}{N_r} \sum_{K=1}^{N_r} (Y_{IJK} - \mu_{IJ})^2 / N_a N_b (N_r - 1).$$

The usual sampling procedure is to select randomly n_a classes from the N_a classes of factor A and independently to select randomly n_b classes from the N_b classes of factor B and from each chosen cell to select a random sample of n_r elements out of N_r elements.

The model equation in this case is the same as (3), that is,

$$y_{ijk} = \mu + a_i + b_j + (ab)_{ij} + e_{ijk}.$$

As before, the random effects a_i , b_j , $(ab)_{ij}$, and e_{ijk} have zero means and are independent of each other. However, they are statistically dependent among themselves with the variance-covariance structure given in Searle and Fawcett (1970). The expected mean squares for this model are shown in Table 5. (See Bennett and Franklin (1954, p. 394), Cornfield and Tukey (1956), Searle and Fawcett (1970)).

TABLE 5
Mean Squares and Expected Mean Squares for the
Finite Model

Source	Mean Squares	Expected Mean Squares
A	S_a^2	$(1 - \frac{n_r}{N_r})\sigma_R^2 + n_r(1 - \frac{n_b}{N_b})\sigma_{AB}^2 + n_b n_r \sigma_A^2$
B	S_b^2	$(1 - \frac{n_r}{N_r})\sigma_R^2 + n_r(1 - \frac{n_a}{N_a})\sigma_{AB}^2 + n_a n_r \sigma_B^2$
AB	S_{ab}^2	$(1 - \frac{n_r}{N_r})\sigma_R^2 + n_r \sigma_{AB}^2$
Residual	S_r^2	σ_R^2

Suppose the usual analysis of variance estimators are calculated under the infinite model while the underlying population is finite and thus the sampling is without replacement. Then the expectations of the variance components estimators (4) are

$$E(\tilde{\sigma}_r^2) = \sigma_R^2, E(\tilde{\sigma}_{ab}^2) = \sigma_{AB}^2 - N_r^{-1} \sigma_R^2,$$

$$E(\tilde{\sigma}_b^2) = \sigma_B^2 - N_a^{-1} \sigma_{AB}^2, \text{ and } E(\tilde{\sigma}_a^2) = \sigma_A^2 - N_b^{-1} \sigma_{AB}^2.$$

Thus if the assumptions of an infinite model are erroneously applied to a finite model situation for the two-way crossed classification design, then the usual analysis of variance estimators $\tilde{\sigma}_a^2$, $\tilde{\sigma}_b^2$, and $\tilde{\sigma}_{ab}^2$ are no longer unbiased and do not necessarily have nonnegative expectations. This is illustrated by a numerical example. Table 6 presents a population for which $E(\tilde{\sigma}_a^2)$ is negative.

TABLE 6
A Finite Two-Way Crossed Classification Population
with Negative $E(\tilde{\sigma}_a^2)$

I \ J		Population Elements Y_{IJK}		
		1	2	3
1	K=1	159	337	49
	K=2	141	345	39
2	K=1	345	441	432
	K=2	331	457	440
3	K=1	384	178	230
	K=2	390	160	200

Here $N_a = 3$, $N_b = 3$, and $N_r = 2$; and the variance components are

$$\sigma_A^2 = \frac{1}{2} \sum_{I=1}^3 (\mu_I - \mu)^2 = 2021.33, \sigma_B^2 = \frac{1}{2} \sum_{J=1}^3 (\mu_J - \mu)^2 = 13580.45,$$

$$\sigma_{AB}^2 = \frac{1}{4} \sum_{I=1}^3 \sum_{J=1}^3 (\mu_{IJ} - \mu_I - \mu_J + \mu)^2 = 16738.33,$$

$$\text{and } \sigma_R^2 = \frac{1}{9} \sum_{I=1}^3 \sum_{J=1}^3 \sum_{K=1}^2 (Y_{IJK} - \mu_{IJ})^2 = 125.78.$$

The expected values of the variance components estimators (4) are

$$E(\tilde{\sigma}_r^2) = \sigma_R^2 = 125.78, E(\tilde{\sigma}_{ab}^2) = \sigma_{AB}^2 - \frac{1}{2}\sigma_R^2 = 1667.44,$$

$$E(\tilde{\sigma}_b^2) = \sigma_B^2 - \frac{1}{3}\sigma_{AB}^2 = 8001.01, \text{ and } E(\tilde{\sigma}_a^2) = \sigma_A^2 - \frac{1}{3}\sigma_{AB}^2 = -3558.11.$$

In general $E(\tilde{\sigma}_a^2) < 0$ if $\sigma_{AB}^2 > N_b \sigma_A^2$, $E(\tilde{\sigma}_b^2) < 0$ if $\sigma_{AB}^2 > N_a \sigma_B^2$, and $E(\tilde{\sigma}_{ab}^2) < 0$ if $\sigma_R^2 > N_r \sigma_{AB}^2$.

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REFERENCES

- ANDERSON, R.L., and BANCROFT, T.A. (1952). *Statistical Theory in Research*. McGraw-Hill Book Co., New York.
- BENNETT, C.A. and FRANKLIN, N.L. (1954). *Statistical Analysis in Chemistry and Chemical Industry*. John Wiley and Sons, New York.
- CORNFIELD, J. and TUKEY, J.W. (1956). Average Values of Mean Squares in Factorials. *Ann. Math. Statist.* **27**, 907-949.
- GRAYBILL, F.A. (1954). On Quadratic Estimates of Variance Components. *Ann. Math. Statist.* **25**, 367-372.
- GRAYBILL, F.A. and HULTQUIST, R.A. (1961). Theorems Concerning Eisenhart's Model II. *Ann. Math. Statist.* **32**, 261-269.
- GRAYBILL, F.A. and WORTHAM, A.W. (1956). A Note on Uniformly Best Unbiased Estimators for Variance Components. *J. Amer. Statist. Ass.* **51**, 266-268.
- McHUGH, R.B. and MIELKE, P.W., Jr. (1968). Negative Variance Estimates and Statistical Dependence in Nested Sampling. *J. Amer. Statist. Ass.*, **63**, 1000-1003.
- SCHEFFÉ, H. (1959). *The Analysis of Variance*. John Wiley and Sons, New York.
- SEARLE, S.R. and FAWCETT, R.F. (1970). Expected Mean Squares in Variance Components Models Having Finite Populations. *Biometrics* **26**, 243-254.
- STEEL, R.G.D. and TORRIE, J.H. (1960). *Principles and Procedures of Statistics*. McGraw-Hill Book Co., New York.

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